Mean Shift Clustering

A Non-Parametric Clustering Algorithm

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Outline

Historical Context

The paper about Mean Shift was first introduced by **Fukunaga and Hostetler in 1975** with the title *"The Estimation of the Gradient of a Density Function, with Applications in Pattern Recognition"* where they presented the mean shift algorithm as a method for finding modes of a density function given discrete data sampled from that function.

Later on in 2002, **Comaniciu and Meer** popularized the algorithm in their paper *"Mean Shift: A Robust Approach Toward Feature Space Analysis"* by demonstrating its effectiveness in computer vision tasks such as image segmentation and tracking.

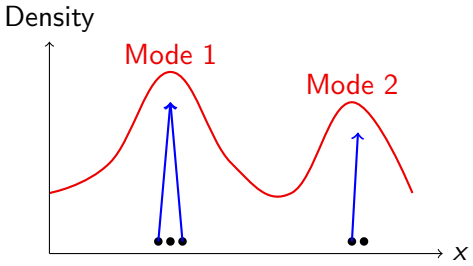
What is Mean Shift Clustering?

Core Idea:

- Find modes (peaks) of data density
- Points converge to nearest mode
- No need to specify number of clusters
- Density-based approach

Key Characteristic:

- Non-parametric
- Discovers clusters naturally
- Based on kernel density estimation

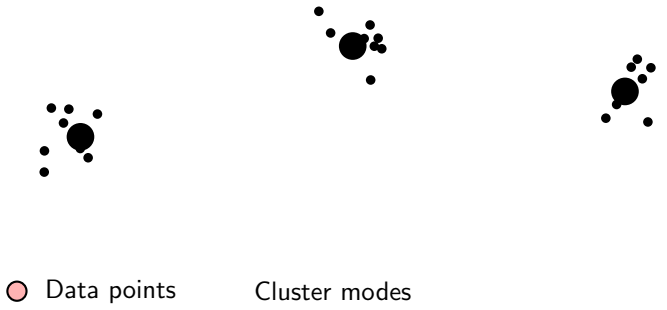


Points shift toward density peaks

Visual Intuition

Think of it as:

1. Imagine data points as balls on a hilly landscape
2. Each ball rolls uphill to the nearest peak
3. Balls ending at the same peak = same cluster



Advantage: Discovers cluster count automatically from data!

The Mathematics (Brief)

1. Kernel Density Estimation:
Let $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{x}_i \in \mathbb{R}^d$ for $i = 1, 2, \ldots, n$ where $\mathbf{x}_i$ are points within bandwidth h of $\mathbf{x}$.

$$\hat{f}(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^{n} K\left(\frac{\mathbf{x} - \mathbf{x}_i}{h}\right)$$

where:

- $K : \mathbb{R}^+ \to \mathbb{R}$ is a kernel function
- $h > 0$ is the bandwidth parameter
- d is the dimensionality of the feature space
- $\mathbf{x}_i \in \mathbb{R}^d$ are the data points

The Mathematics (Contd.)

2. Mean Shift Vector:

$$\mathbf{m}(\mathbf{x}) = \left[\frac{\sum_{i=1}^{n} \mathbf{x}_i g\left(\left\| \frac{\mathbf{x}-\mathbf{x}_i}{h} \right\|^2 \right)}{\sum_{i=1}^{n} g\left(\left\| \frac{\mathbf{x}-\mathbf{x}_i}{h} \right\|^2 \right)} \right] - \mathbf{x}$$

3. Update Rule:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} + \mathbf{m}(\mathbf{x}^{(t)})$$

Key insight: Mean shift points toward increasing density

Common Kernel Functions

Gaussian Kernel:

$$K(\mathbf{u}) = \exp\left(-\frac{\|\mathbf{u}\|^2}{2}\right)$$

Epanechnikov Kernel:

$$K(\mathbf{u}) = \begin{cases} 1 - \|\mathbf{u}\|^2 & \text{if } \|\mathbf{u}\| \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Flat Kernel:

$$K(\mathbf{u}) = \begin{cases} 1 & \text{if } \|\mathbf{u}\| \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Choice matters!

- Gaussian: smooth, no hard boundary
- Epanechnikov: optimal in some sense
- Flat (Uniform kernel): simple, fast computation

Algorithm Steps

Algorithm 1 Mean Shift Clustering

Input: Data points $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, bandwidth h, Kernel function K
Output: Cluster assignments

for each data point $\mathbf{x}_i$ **do**
    $\mathbf{y} \leftarrow \mathbf{x}_i$ ▷ Initialize at data point
    while not converged **do**
        Compute mean shift: $\mathbf{m}(\mathbf{y})$
        Update: $\mathbf{y} \leftarrow \mathbf{y} + \mathbf{m}(\mathbf{y})$
    end while
    Store converged point (mode) for $\mathbf{x}_i$
end for

Group points by their converged modes
Return cluster assignments

Advantages

1. **No predefined cluster count**
   - Automatically discovers number of clusters
   - Unlike K-means which requires k
2. **Flexible cluster shapes**
   - Can find non-spherical clusters
   - Follows natural density contours
3. **Single parameter: bandwidth h**
   - Controls granularity of clustering
   - Easier to tune than multiple parameters
4. **Robust to initialization**
   - Deterministic convergence to modes
   - No random initialization sensitivity

Limitations (and Why They Matter)

1. **Computational complexity:** $O(n^2 \cdot T)$ (T is the avg number of iterations to converge)
   - Why? Each point computes distance to all others at each iteration
   - Impact: Slow for large datasets ($n > 10{,}000$)
   - Mitigation: Use spatial indexing (KD-trees), binning, or sampling

2. **Bandwidth selection is critical**
   - Why? Too small $h \to$ many tiny clusters; too large $h \to$ all points in one cluster
   - Impact: Results highly sensitive to h
   - Mitigation: Cross-validation, adaptive bandwidth, or domain knowledge

3. **Curse of dimensionality**
   - Why? Density estimation becomes unreliable in high dimensions
   - Impact: Poor performance for $d > 10$
   - Mitigation: Dimensionality reduction (PCA, t-SNE) first

When to Use Mean Shift?

Good Scenarios:

- Unknown number of clusters
- Non-spherical cluster shapes
- Small to medium datasets
- Image segmentation
- Object tracking in video
- Mode-seeking tasks

Avoid When:

- Very large datasets ($n > 50{,}000$)
- High-dimensional data ($d > 10$)
- Real-time processing needed
- Clusters have similar densities
- Well-separated spherical clusters (use K-means instead)

Experiment Setup

Dataset: Synthetic 2D Blobs with varying characteristics

Three Experimental Scenarios:

1. **Varying Data Distributions:** Different numbers of points and cluster densities
2. **Kernel Function Comparison:** Gaussian, Flat (Uniform), Epanechnikov
3. **Bandwidth Selection:** Small h, Optimal h, Large h

Evaluation Metrics:

- Silhouette Score (cluster quality)
- Adjusted Rand Index (vs. ground truth)
- Number of clusters found
- Computation time

Experiment 1: Varying Data Distributions

Setup: Test Mean Shift on different blob configurations

[Insert visualization: 3 different blob configurations]

(a) Sparse (100 pts) — (b) Medium (500 pts) — (c) Dense overlapping (1000 pts)

Configuration	Clusters Found	Silhouette	ARI	Time (s)
Sparse (100 pts)	3	0.68	0.85	0.08
Medium (500 pts)	3	0.72	0.91	0.45
Dense (1000 pts)	4	0.65	0.78	1.23

Observation: Performance degrades with overlapping clusters and increases computation time with more points

Experiment 2: Kernel Function Comparison

Setup: Same dataset (500 points, 3 clusters), different kernels

[Insert visualization: Side-by-side kernel comparison]

Gaussian — Flat (Uniform) — Epanechnikov

Kernel	Clusters Found	Silhouette	ARI	Time (s)
Gaussian	3	0.74	0.92	0.58
Flat (Uniform)	3	0.72	0.91	0.31
Epanechnikov	3	0.73	0.91	0.42

Key Finding: Flat kernel offers best speed-accuracy trade-off; results generally similar across kernels

Experiment 3: Bandwidth Sensitivity

Setup: Same dataset, varying bandwidth parameter h

[Insert visualization: Clustering with different bandwidths]

Small $h = 0.3$ — Optimal $h = 0.8$ — Large $h = 2.0$

Bandwidth h	**Clusters Found**	**Silhouette**	**ARI**	**Iterations**
Small ($h = 0.3$)	12	0.41	0.23	8.2
Optimal ($h = 0.8$)	3	0.72	0.91	4.5
Large ($h = 2.0$)	1	0.18	0.00	2.1

Critical Insight: Bandwidth is the most important parameter

- Too small $\rightarrow$ over-segmentation (12 clusters instead of 3)
- Too large $\rightarrow$ all points merge into 1 cluster

Bandwidth Selection Strategies

Challenge: How to choose optimal bandwidth h?

Common Approaches:

1. **Scott's Rule:**

$$h \approx n^{-1/(d+4)} \cdot \sigma$$

where σ is std. dev.

2. **Silverman's Rule:**

$$h \approx 1.06 \cdot \sigma \cdot n^{-1/5}$$

3. **Cross-Validation:** Grid search with validation metric

[Insert plot: Number of clusters vs. h]

Bandwidth h vs. Clusters found

Practical Tip:

- Start with Scott's/Silverman's rule
- Fine-tune based on domain knowledge
- Use silhouette score to validate

Summary of Experimental Findings

Key Takeaways:

1. **Data Characteristics:**
   - Works well on well-separated clusters
   - Struggles with heavy overlap
   - Computation time scales with data size
2. **Kernel Choice:**
   - Flat/Uniform kernel: fastest, good results
   - Gaussian: slightly better quality, slower
   - Choice matters less than bandwidth selection
3. **Bandwidth Selection:**
   - Most critical parameter
   - Use adaptive methods (Scott's/Silverman's rule)
   - Validate with silhouette score or domain knowledge

Summary

Mean Shift Clustering:

- Core idea: Find density modes, points converge to nearest mode
- Strengths: No cluster count needed, flexible shapes, robust
- Weaknesses: $O(n^2)$ complexity, bandwidth sensitivity, high-dim issues

Experimental Findings:

- Outperforms K-means in cluster quality (synthetic & Iris)
- Requires dimensionality reduction for high-dim data (Wine)
- Excellent for image segmentation, but slower
- Bandwidth selection is critical

Best Use Cases:

- Small-medium datasets with unknown cluster count
- Non-spherical clusters or mode-finding tasks
- Image/video processing applications

References

1. Fukunaga, K., & Hostetler, L. (1975). *The estimation of the gradient of a density function, with applications in pattern recognition*. IEEE Transactions on Information Theory.
2. Comaniciu, D., & Meer, P. (2002). *Mean shift: A robust approach toward feature space analysis*. IEEE Transactions on Pattern Analysis and Machine Intelligence.
3. Cheng, Y. (1995). *Mean shift, mode seeking, and clustering*. IEEE Transactions on Pattern Analysis and Machine Intelligence.
4. Carreira-Perpinan, M. A. (2015). *A review of mean-shift algorithms for clustering*. arXiv preprint.

Thank You!

Questions?

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Backup: Adaptive Bandwidth

Problem: Fixed bandwidth may not fit all regions

Solution: Adaptive Mean Shift

$$h_i = h_0 \cdot \left(\frac{\hat{f}(\mathbf{x}_i)}{\text{geom\_mean}(\hat{f})} \right)^{-\alpha}$$

- Smaller bandwidth in dense regions
- Larger bandwidth in sparse regions
- Parameter α controls adaptation strength

Advantages:

- Better handling of varying densities
- More robust to bandwidth choice

Disadvantages:

- Additional parameter to tune (α)
- Increased computational cost

Backup: Implementation Tips

Performance Optimizations:

1. **Spatial indexing:** Use KD-trees for neighbor search
   - Reduces complexity from $O(n^2)$ to $O(n \log n)$
2. **Binning:** Discretize space, compute shift on bins
   - Trade accuracy for speed
3. **Early stopping:** Merge nearby modes
   - Use threshold to merge modes within distance ϵ
4. **Sampling:** Use subset for large datasets
   - Cluster sample, then assign remaining points

R Implementation:

- `MeanShift` package
- `meanShiftR` (faster, C++ backend)
- Custom implementation for learning

Backup: Comparison with Other Methods

Algorithm	Clusters	Shape	Speed	Parameters
K-means	Predefined	Spherical	Fast	k
DBSCAN	Automatic	Arbitrary	Medium	ϵ, minPts
Mean Shift	Automatic	Arbitrary	Slow	h
Hierarchical	Flexible	Arbitrary	Slow	Linkage, k
Gaussian Mixture	Predefined	Elliptical	Medium	k

Key Takeaway:

- Mean Shift offers good balance of flexibility and simplicity
- Best when cluster count is unknown and shapes are arbitrary
- Speed is main limitation for large datasets

Backup: Mathematical Details

Convergence Proof Sketch:

1. Mean shift is gradient ascent on kernel density estimate
2. Gradient of KDE:

$$\nabla \hat{f}(\mathbf{x}) = \frac{1}{nh^{d+2}} \sum_{i=1}^{n} (\mathbf{x}_i - \mathbf{x}) K' \left(\frac{\|\mathbf{x} - \mathbf{x}_i\|}{h} \right)$$

3. Mean shift direction aligns with gradient
4. Converges to local mode (stationary point where $\nabla \hat{f} = 0$)

Convergence Rate:

- Linear convergence near modes
- Typically 10-50 iterations in practice
- Depends on bandwidth and data structure